CHMENG 250: Transport Processes

Homework 2: Kinematics of Fluid Motion

Assigned: Fri, Sep. 4, 10:00 a.m. (PT) **Due:** Sat, Sep. 19, 11:59 p.m. (PT)

Convention. Unless stated otherwise, all indices range over $1, 2, 3$, repeated indices are summed. Cartesian components refer to a fixed orthonormal basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$. Throughout, the velocity gradient is

$$(\boldsymbol{\nabla v})_{ij} = \partial_j v_i,$$

so that the row index names the velocity component and the column index the direction of differentiation. Several textbooks use the opposite convention; check indices rather than the placement of $\boldsymbol{\nabla}$ before importing a formula.

Problem 1: Differential identities common in transport phenomena [12 points]

Let ϕ be a scalar field and let $\boldsymbol{v}$ and $\boldsymbol{w}$ be vector fields on $\mathbb{R}^3$. Prove each identity using indicial notation. In every case, state which property of δ_{ij}, of ϵ_{ijk}, or of the commuting partial derivatives $\partial_i \partial_j = \partial_j \partial_i$ does the essential work.

(a) The two product rules

$$\begin{aligned} \boldsymbol{\nabla} \cdot (\phi \boldsymbol{v}) &= \phi \boldsymbol{\nabla} \cdot \boldsymbol{v} + \boldsymbol{v} \cdot \boldsymbol{\nabla} \phi, \\ \boldsymbol{\nabla} \times (\phi \boldsymbol{v}) &= \phi \boldsymbol{\nabla} \times \boldsymbol{v} + \boldsymbol{\nabla} \phi \times \boldsymbol{v}. \end{aligned}$$ [2 points]

(b) The gradient of a dot product,

$$\boldsymbol{\nabla} (\boldsymbol{v} \cdot \boldsymbol{w}) = (\boldsymbol{\nabla v})^{\top} \boldsymbol{w} + (\boldsymbol{\nabla w})^{\top} \boldsymbol{v}.$$ [2 points]

(c) The divergence of a dyad,

$$\boldsymbol{\nabla} \cdot (\boldsymbol{v} \otimes \boldsymbol{w}) = (\boldsymbol{\nabla v})\, \boldsymbol{w} + (\boldsymbol{\nabla} \cdot \boldsymbol{w})\, \boldsymbol{v}.$$ [2 points]

This is the identity that splits the convective momentum flux $\rho \boldsymbol{v} \otimes \boldsymbol{v}$ into its two parts in Problem 8.

(d) The divergence of a cross product,

$$\boldsymbol{\nabla} \cdot (\boldsymbol{v} \times \boldsymbol{w}) = \boldsymbol{w} \cdot \boldsymbol{\nabla} \times \boldsymbol{v} - \boldsymbol{v} \cdot \boldsymbol{\nabla} \times \boldsymbol{w}.$$ [2 points]

(e) The curl of a curl,

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{v}) = \boldsymbol{\nabla} (\boldsymbol{\nabla} \cdot \boldsymbol{v}) - \nabla^2 \boldsymbol{v}.$$

Deduce that in an incompressible flow the viscous force per unit volume may be written as

$$\mu \nabla^2 \boldsymbol{v} = -\mu\, \boldsymbol{\nabla} \times \boldsymbol{\omega}, \qquad \boldsymbol{\omega} \coloneqq \boldsymbol{\nabla} \times \boldsymbol{v},$$

and say in one sentence what this form asserts about the direction in which momentum diffuses. [2 points]

(f) The identity

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \boldsymbol{v})^{\top} = \boldsymbol{\nabla} (\boldsymbol{\nabla} \cdot \boldsymbol{v}),$$

and hence, with $\boldsymbol{D} \coloneqq \frac{1}{2}\left[\boldsymbol{\nabla} \boldsymbol{v} + (\boldsymbol{\nabla} \boldsymbol{v})^{\top}\right]$,

$$\boldsymbol{\nabla} \cdot (2\boldsymbol{D}) = \nabla^2 \boldsymbol{v} + \boldsymbol{\nabla} (\boldsymbol{\nabla} \cdot \boldsymbol{v}).$$

State what this reduces to for an incompressible flow. [2 points]

Problem 2: Symmetry and the stress power [9 points]

Let $\boldsymbol{T}$ and $\boldsymbol{S}$ be second-rank tensors on $\mathbb{R}^3$, and recall the double contraction $\boldsymbol{T} : \boldsymbol{S} = T_{ij} S_{ij}$. Recall also the two symmetry classes:

$$\begin{aligned} \boldsymbol{T} \text{ is } \textit{symmetric} &\iff \boldsymbol{T} = \boldsymbol{T}^{\top} \iff T_{ij} = T_{ji}, \\ \boldsymbol{T} \text{ is } \textit{antisymmetric} &\iff \boldsymbol{T} = -\boldsymbol{T}^{\top} \iff T_{ij} = -T_{ji}. \end{aligned}$$

The terms *antisymmetric* and *skew-symmetric* are used interchangeably. An antisymmetric tensor has $T_{ii} = -T_{ii}$ for each i, hence vanishing diagonal entries, leaving only three independent components in three dimensions.

(a) Show that every $\boldsymbol{T}$ decomposes uniquely into a symmetric and an antisymmetric part,

$$\boldsymbol{T} = \operatorname{sym} \boldsymbol{T} + \operatorname{skw} \boldsymbol{T}, \qquad \operatorname{sym} \boldsymbol{T} \coloneqq \tfrac{1}{2}\left(\boldsymbol{T} + \boldsymbol{T}^{\top}\right), \qquad \operatorname{skw} \boldsymbol{T} \coloneqq \tfrac{1}{2}\left(\boldsymbol{T} - \boldsymbol{T}^{\top}\right),$$

and that $\boldsymbol{A} : \boldsymbol{B} = 0$ whenever $\boldsymbol{A}$ is symmetric and $\boldsymbol{B}$ is antisymmetric. [3 points]

(b) Suppose $\boldsymbol{T} : \boldsymbol{S} = 0$ for *every* antisymmetric $\boldsymbol{S}$. Prove that $\boldsymbol{T}$ is symmetric. [2 points]

Hint

Choose $\boldsymbol{S}$ to be the antisymmetric part of $\boldsymbol{T}$ itself.

(c) Let $\boldsymbol{L} \coloneqq \boldsymbol{\nabla} \boldsymbol{v}$ with $\boldsymbol{L} = \boldsymbol{D} + \boldsymbol{W}$, where $\boldsymbol{D} = \operatorname{sym} \boldsymbol{L}$ is the rate-of-strain tensor and $\boldsymbol{W} = \operatorname{skw} \boldsymbol{L}$ is the rotation tensor. Show that for any *symmetric* $\boldsymbol{T}$,

$$\boldsymbol{T} : \boldsymbol{L} = \boldsymbol{T} : \boldsymbol{D}.$$

The quantity $\boldsymbol{T} : \boldsymbol{\nabla v}$ is the rate at which stresses do work per unit volume. Explain in one or two sentences what your result says about the ability of the local rotation $\boldsymbol{W}$ to do work, and why that is what one should expect on physical grounds. [2 points]

(d) Prove the product rule

$$\boldsymbol{\nabla} \cdot \left(\boldsymbol{T}^{\top} \boldsymbol{v} \right) = (\boldsymbol{\nabla} \cdot \boldsymbol{T}) \cdot \boldsymbol{v} + \boldsymbol{T} : \boldsymbol{\nabla v},$$

valid for any second-rank tensor field $\boldsymbol{T}$. Identify which term becomes the divergence of a mechanical-energy flux and which becomes the local working of the stress. [2 points]

Problem 3: Anatomy of a linear planar flow [14 points]

Consider the steady velocity field

$$\boldsymbol{v} = (\alpha x_1 + \gamma x_2)\, \boldsymbol{e}_1 + (\beta x_1 - \alpha x_2)\, \boldsymbol{e}_2,$$

with α, β, and γ real constants. This single family contains simple shear, pure straining, and solid-body rotation as special cases.

The following definitions are the ones to use, in the forms given. A flow is *incompressible* (isochoric) when every material element preserves its volume, which by the volume theorem of Chapter 2 means

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \operatorname{tr} \boldsymbol{L} = 0.$$

It is *irrotational* when the vorticity vanishes,

$$\boldsymbol{\omega} \coloneqq \boldsymbol{\nabla} \times \boldsymbol{v} = \boldsymbol{0} \quad \Longleftrightarrow \quad \boldsymbol{W} = \boldsymbol{0},$$

in which case the motion is a *pure straining* flow. A *simple shear* is a flow in which one velocity component varies linearly in one transverse direction and nothing else happens,

$$\boldsymbol{v} = \dot{\gamma}\, x_2\, \boldsymbol{e}_1,$$

and a *solid-body rotation* is one in which the fluid turns rigidly at angular velocity $\boldsymbol{\Omega}$,

$$\boldsymbol{v} = \boldsymbol{\Omega} \times \boldsymbol{x} \quad \Longleftrightarrow \quad \boldsymbol{D} = \boldsymbol{0}.$$

(a) Write down $\boldsymbol{L} = \boldsymbol{\nabla v}$ and its symmetric and antisymmetric parts $\boldsymbol{D}$ and $\boldsymbol{W}$. Show that the flow is incompressible for every choice of the three constants. [3 points]

(b) Compute the vorticity $\boldsymbol{\omega} = \boldsymbol{\nabla} \times \boldsymbol{v}$ and the local angular velocity $\boldsymbol{w} = \frac{1}{2}\boldsymbol{\omega}$. Determine the conditions on α, β, and γ under which the flow is

(i) irrotational, (ii) a simple shear, (iii) a solid-body rotation.

In each case say what becomes of $\boldsymbol{D}$ and of $\boldsymbol{W}$. [3 points]

(c) Find the principal strain rates, that is, the eigenvalues of $\boldsymbol{D}$. Verify that they sum to zero and explain why they must, given your answer to Problem 3(a). [3 points]

(d) A material fiber has instantaneous unit direction $\boldsymbol{m} = \cos\theta\,\boldsymbol{e}_1 + \sin\theta\,\boldsymbol{e}_2$. Show that its fractional rate of stretching is

$$\frac{1}{\ell}\frac{\mathrm{d}\ell}{\mathrm{d}t} = \boldsymbol{m}\cdot\boldsymbol{D}\cdot\boldsymbol{m} = \alpha\cos 2\theta + \frac{\beta+\gamma}{2}\sin 2\theta,$$

and find the orientation θ that maximizes it. Confirm that the maximum equals the largest principal strain rate found in Problem 3(c). [2 points]

(e) Compute the material acceleration $\boldsymbol{a} = \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$ directly, and verify that it agrees with the form

$$\boldsymbol{a} = \boldsymbol{\nabla}\left(\tfrac{1}{2}\,|\boldsymbol{v}|^2\right) - \boldsymbol{v}\times\boldsymbol{\omega}$$

supplied by the BAC–CAB rule, proved in Chapter 1 and covered in Lecture 2. The field is steady; explain why the acceleration is nevertheless nonzero, and identify the one combination of α, β, and γ on which it depends. [3 points]

Problem 4: Operators in cylindrical and spherical coordinates [12 points]

Every identity in Problem 1 was written in a fixed Cartesian basis. In curvilinear coordinates the unit vectors rotate from point to point, so differentiating a field means differentiating the basis vectors as well as the components, and extra algebraic terms appear. The resulting formulas are collected in Appendix 1.A; quote them here rather than rederiving them. Cylindrical coordinates are (r,θ,z); spherical coordinates are (r,θ,ϕ) with θ the polar angle and ϕ the azimuthal angle. Both are examples of orthogonal curvilinear coordinates, in that the unit vectors are orthonormal.

(a) *Cylindrical gradient and Laplacian.* For $f = A\ln r + Bz$ with A and B constant, compute $\boldsymbol{\nabla} f$ and $\nabla^2 f$. Show that f is harmonic ($\nabla^2 f = 0$) everywhere except on the axis $r = 0$. [2 points]

(b) *Cylindrical divergence and curl, radial field.* A line source on the axis produces

$$\boldsymbol{v} = \frac{Q}{2\pi r}\,\boldsymbol{e}_r,$$

with Q constant. Compute $\boldsymbol{\nabla}\cdot\boldsymbol{v}$ and $\boldsymbol{\nabla}\times\boldsymbol{v}$ for $r > 0$. Both vanish, yet fluid streams steadily outward; reconcile this by computing the volumetric flux $\oint \boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}A$ through a cylinder of radius R and unit height, and noting how it depends on R. [2 points]

(c) *Cylindrical curl, azimuthal field.* For a purely azimuthal field $\boldsymbol{v} = v_\theta(r)\,\boldsymbol{e}_\theta$, show that the only surviving component of the vorticity is

$$\omega_z = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\,v_\theta\right).$$

Among the power laws $v_\theta = Cr^n$, find the exponent n that makes the flow irrotational for $r > 0$, and the exponent that gives uniform vorticity. [2 points]

(d) *Spherical gradient and Laplacian.* For $f = A/r + Br\cos\theta$ with A and B constant, compute $\boldsymbol{\nabla} f$ and show that $\nabla^2 f = 0$ for $r > 0$. Identify each of the two terms physically. [2 points]

(e) *Spherical divergence.* For the point-source field $\boldsymbol{v} = (C/r^2)\,\boldsymbol{e}_r$, show that $\boldsymbol{\nabla}\cdot\boldsymbol{v} = 0$ for $r > 0$, and compute the flux through a sphere of radius R. Comment on its dependence on R. [2 points]

(f) *Spherical curl.* A fluid in solid-body rotation at angular velocity Ω about the polar axis has $v_\phi = \Omega r\sin\theta$ and $v_r = v_\theta = 0$. Compute $\boldsymbol{\nabla}\times\boldsymbol{v}$ and verify that it equals $2\Omega\,\boldsymbol{e}_z$, using $\boldsymbol{e}_z = \cos\theta\,\boldsymbol{e}_r - \sin\theta\,\boldsymbol{e}_\theta$. [2 points]

Problem 5: Pathlines, streamlines, and what a particle experiences [13 points]

Two families of curves are used to visualize a flow. A *streamline* is a curve everywhere tangent to the velocity field at one *fixed* instant; holding t fixed, it is obtained by integrating

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}s} = \boldsymbol{v}(\boldsymbol{x}, t), \qquad \text{equivalently} \qquad \frac{\mathrm{d}x_1}{v_1} = \frac{\mathrm{d}x_2}{v_2} = \frac{\mathrm{d}x_3}{v_3}.$$

A *pathline* is the trajectory actually traced out by one material particle over time, obtained by integrating

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{v}\left(\boldsymbol{x}(t), t\right), \qquad \boldsymbol{x}(0) = \boldsymbol{x}_0.$$

The two answer different questions, and in general they are different curves.

Consider the unsteady flow with the spatially uniform velocity field

$$\boldsymbol{v} = u_0\,\boldsymbol{e}_1 + kt\,\boldsymbol{e}_2,$$

with $u_0 > 0$ and $k > 0$ constant.

(a) Find the streamlines at a fixed instant t and sketch the family for two different times. [3 points]

(b) Find the pathline of the particle that occupies $\boldsymbol{x}_0 = x_{10}\boldsymbol{e}_1 + x_{20}\boldsymbol{e}_2$ at $t = 0$. Integrate $\mathrm{d}x_1/\mathrm{d}t = u_0$ and $\mathrm{d}x_2/\mathrm{d}t = kt$ separately subject to that initial condition, then eliminate t between the two results to obtain the trajectory in the x_1x_2-plane. Sketch it, and say what curve it is. [4 points]

(c) The two families are different curves. State the condition under which they coincide, and explain in one or two sentences why that condition is the right one. [2 points]

(d) Compute $\boldsymbol{L}$, $\boldsymbol{D}$, $\boldsymbol{W}$, and $\boldsymbol{\omega}$ for this flow, and then compute the material acceleration $\boldsymbol{a} = \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$. A fluid particle is accelerating even though no material element is deforming or spinning; reconcile the two statements. Finally, for the steady temperature field $T = T_0 + \lambda x_2$ carried by this flow,

where the constant $\lambda > 0$, evaluate $\frac{\partial T}{\partial t}$ and $\frac{\mathrm{D}T}{\mathrm{D}t}$, and contrast the result with a flow in which the roles of the local and convective terms are reversed. [4 points]

Problem 6: How material elements stretch [13 points]

[Advanced] Let $\Delta \boldsymbol{r} = \ell\, \boldsymbol{m}$ be an infinitesimal material line element, with length $\ell = |\Delta \boldsymbol{r}|$ and unit direction $\boldsymbol{m}$, convected by the flow so that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\Delta \boldsymbol{r}\right) = \boldsymbol{L} \cdot \Delta \boldsymbol{r}, \qquad \boldsymbol{L} = \boldsymbol{\nabla} \boldsymbol{v}.$$

(a) Derive

$$\frac{1}{\ell}\frac{\mathrm{d}\ell}{\mathrm{d}t} = \boldsymbol{m} \cdot \boldsymbol{D} \cdot \boldsymbol{m},$$

and explain precisely where the rotation tensor $\boldsymbol{W}$ drops out. [4 points]

(b) Show that the direction itself obeys

$$\frac{\mathrm{d}\boldsymbol{m}}{\mathrm{d}t} = \boldsymbol{L} \cdot \boldsymbol{m} - \left(\boldsymbol{m} \cdot \boldsymbol{D} \cdot \boldsymbol{m}\right) \boldsymbol{m},$$

and verify that $\boldsymbol{m} \cdot \mathrm{d}\boldsymbol{m}/\mathrm{d}t = 0$, so that $\boldsymbol{m}$ remains a unit vector. [3 points]

(c) Label each fluid particle by the position $\boldsymbol{x}_0$ it occupied at $t = 0$, the *reference configuration*, and let $\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{x}_0, t)$ be the position of that same particle at time t, the *current configuration*. Collect the derivatives of the current position with respect to the reference position in the array

$$F_{ij} \coloneqq \frac{\partial \chi_i}{\partial x_{0j}},$$

and write $J \coloneqq \det \boldsymbol{F}$ for its determinant (det denotes the determinant). A small material element occupying volume $\mathrm{d}V_0$ at $t = 0$ occupies $\mathrm{d}V = J\, \mathrm{d}V_0$ at time t, so J is precisely the factor by which the element has expanded.

Show first that $\mathrm{d}\boldsymbol{F}/\mathrm{d}t = \boldsymbol{L} \cdot \boldsymbol{F}$, and then, using Jacobi's formula

$$\frac{\mathrm{d}}{\mathrm{d}t} \det \boldsymbol{A} = \left(\det \boldsymbol{A}\right) \operatorname{tr}\left(\boldsymbol{A}^{-1} \frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t}\right) \qquad \text{for invertible } \boldsymbol{A}(t),$$

prove Euler's expansion formula

$$\frac{\mathrm{d}J}{\mathrm{d}t} = \left(\boldsymbol{\nabla} \cdot \boldsymbol{v}\right) J.$$

Since $\mathrm{d}V_0$ is fixed once the particle has been labelled, conclude that

$$\frac{1}{\mathrm{d}V}\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{d}V\right) = \boldsymbol{\nabla} \cdot \boldsymbol{v}.$$ [3 points]

(d) Deduce that an incompressible flow has $\operatorname{tr} \boldsymbol{D} = 0$, so that its principal strain rates sum to zero. Apply this to simple shear $\boldsymbol{v} = \dot{\gamma} x_2 \, \boldsymbol{e}_1$, whose principal strain rates are $\pm \dot{\gamma}/2$, and describe in words what happens to a small material disk carried by the flow. [3 points]

Problem 7: The Rankine vortex [12 points]

A useful idealization of a real vortex, due to W. J. M. Rankine, rotates like a solid body inside a core of radius a and like a point vortex outside it. In cylindrical coordinates (r, θ, z),

$$v_r = v_z = 0, \qquad v_\theta(r) = \begin{cases} \Omega r, & r \le a, \\ \Omega a^2 / r, & r > a, \end{cases}$$

with $\Omega > 0$ constant. The velocity is continuous at $r = a$. These are exactly the two power laws identified in Problem 4(c).

(a) Verify that $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$ everywhere, and state whether this required the specific form of $v_\theta(r)$. [2 points]

(b) Compute the vorticity $\boldsymbol{\omega} = \boldsymbol{\nabla} \times \boldsymbol{v}$ in both regions. Comment on the fact that the streamlines are circles of the same shape in both regions while the vorticity is not the same. [3 points]

(c) Compute the circulation

$$\Gamma(R) = \oint_{C_R} \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{\ell}$$

around the circle C_R of radius R centered on the axis, for $R \le a$ and for $R > a$. Verify Stokes' theorem in each case, and identify the value of R beyond which Γ stops growing. [3 points]

(d) Compute the rate-of-strain tensor $\boldsymbol{D}$ in both regions. Show that the core is a pure rotation with no deformation whatsoever, while the exterior deforms fluid elements without spinning them. [2 points]

(e) With the convention

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}, \qquad v_\theta = -\frac{\partial \psi}{\partial r},$$

find a stream function $\psi(r)$ for the flow, choosing the constants so that ψ is continuous at $r = a$ and $\psi(0) = 0$. [2 points]

Problem 8: Transport, mass, and traction

This problem collects the opening results of Chapter 3. Throughout, $V(t)$ is a material volume with boundary $S(t)$ and outward unit normal $\boldsymbol{n}$, and $\rho(\boldsymbol{x}, t)$ is the mass density.

(a) Let ψ be any property per unit mass. Starting from the Reynolds transport theorem applied to

$\phi = \rho\psi$, and using conservation of mass, prove the per-unit-mass form

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V(t)} \rho\,\psi \,\mathrm{d}V = \int_{V(t)} \rho\,\frac{\mathrm{D}\psi}{\mathrm{D}t}\,\mathrm{d}V.$$

Point out exactly where continuity is used. [3 points]

(b) Let $V_{\mathrm{CV}}(t)$ be a control volume whose boundary moves with a velocity $\boldsymbol{w}$ of your choosing, and let it coincide with $V(t)$ at the instant considered. Derive

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V(t)} \phi\,\mathrm{d}V = \frac{\mathrm{d}}{\mathrm{d}t}\int_{V_{\mathrm{CV}}(t)} \phi\,\mathrm{d}V + \oint_{S_{\mathrm{CV}}(t)} \phi\,(\boldsymbol{v}-\boldsymbol{w})\cdot\boldsymbol{n}\,\mathrm{d}A.$$

Specialize to a fixed control volume in steady flow with $\phi = \rho$, and obtain the familiar statement $\rho_1 U_1 A_1 = \rho_2 U_2 A_2$ for a stream tube with one inlet and one outlet. [3 points]

(c) Write the continuity equation in cylindrical coordinates, and reduce it for an incompressible axisymmetric flow with $v_\theta = 0$. Then verify that

$$v_r = -\tfrac{1}{2}ar, \qquad\qquad v_z = az,$$

with a constant, satisfies it, and describe the flow this field represents. [3 points]

(d) At a point of a fluid the stress tensor has Cartesian components, in pascals,

$$\boldsymbol{T} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 3 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Let $\boldsymbol{n} = (\boldsymbol{e}_1 + \boldsymbol{e}_2 + \boldsymbol{e}_3)/\sqrt{3}$.

(i) Compute the traction $\boldsymbol{t} = \boldsymbol{T}\boldsymbol{n}$.

(ii) Resolve $\boldsymbol{t}$ into its normal and shear parts and give the magnitude of each.

(iii) Verify that $\boldsymbol{t}(-\boldsymbol{n}) = -\boldsymbol{t}(\boldsymbol{n})$, and name the result this illustrates.

(iv) Find the mean normal stress $\frac{1}{3}\operatorname{tr}\boldsymbol{T}$ and the deviatoric stress. [3 points]

(e) The balance of linear momentum for a material volume reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V(t)} \rho\,\boldsymbol{v}\,\mathrm{d}V = \oint_{S(t)} \boldsymbol{t}\,\mathrm{d}A + \int_{V(t)} \rho\,\boldsymbol{b}\,\mathrm{d}V,$$

with $\boldsymbol{b}$ the body force per unit mass. Using Problem 8(a), Cauchy's stress theorem $\boldsymbol{t} = \boldsymbol{T}\boldsymbol{n}$, the divergence theorem, and the localization argument, derive the local form

$$\rho\,\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} = \boldsymbol{\nabla}\cdot\boldsymbol{T} + \rho\,\boldsymbol{b}.$$

For each of the four ingredients, state whether it is a universal balance law, a consequence of one, a definition, or a mathematical identity. [3 points]